

Beyond the Image: Enhancing Movie Genre Classification with Title-Aware Cross-Attention on Poster Features

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Abstract—Accurate movie genre classification underpins key functionalities in modern streaming platforms, including recommendation systems, content filtering, and metadata management. Movie posters offer a lightweight, universally available visual signal for prerelease genre prediction; however, visual features alone often fail to disambiguate genres that share similar aesthetic conventions. This paper investigates whether incorporating the movie title as a complementary textual signal can improve multilabel genre classification from poster images. We propose a multimodal architecture that fuses EfficientNet-B3 visual features with BERT-encoded title embeddings via an 8-head cross-attention mechanism, and compare it against an image-only EfficientNet-B3 baseline under identical training conditions. Both models are evaluated on a dataset of 300 IMDb poster images covering 14 genre categories. The multimodal model achieves a Macro F1 of 0.7293 after per-genre threshold optimization, outperforming the image-only baseline by +0.2024 in Macro F1. The performance gain is most pronounced on lexically distinctive genres such as *sci-fi*, *horror*, and *adventure*, where title semantics guide the cross-attention mechanism to attend to genre-discriminative visual regions, as confirmed by GradCAM visualizations. These results demonstrate that movie titles, despite their brevity, encode genre-discriminative information that meaningfully complements visual poster features for multilabel genre classification.

Index Terms—movie genre classification, EfficientNet, BERT, cross-attention, multilabel classification, movie poster

Code and Notebook Availability: The source code and training notebooks used in this report are publicly available at <https://github.com/nafalrust/DIPandCompViProject-MovieGenreClassification.git>.

I. INTRODUCTION

The rapid growth of streaming platforms and digital content distribution has generated an unprecedented volume of movies accessible to global audiences. Major platforms such as Netflix, Amazon Prime, and Disney+ now host tens of thousands of titles spanning a wide variety of genres, mak-

ing efficient content organization and discovery increasingly critical. Accurate genre labels underpin key functionalities including recommendation systems, content filtering, audience targeting, and metadata management. While genre annotations are typically assigned manually by human editors, the sheer scale of modern content libraries makes fully manual labeling impractical, motivating the need for automated genre classification systems.

Movie posters serve as one of the earliest and most accessible visual representations of a film, encoding rich semantic cues such as color palettes, typography, facial expressions, and compositional elements that often reflect the underlying genre. A horror film poster, for instance, tends to employ dark tones and unsettling imagery, whereas an animation poster typically features vibrant colors and stylized characters. Unlike trailers, subtitles, or plot synopses, posters are publicly available prior to a film’s release, making them a particularly valuable modality for prerelease genre prediction. Furthermore, poster images are lightweight and universally accessible, requiring no transcription, translation, or temporal processing, which further strengthens their appeal as a primary input modality.

Movie genre classification from poster images is inherently a multilabel problem, as a single film can simultaneously belong to multiple genres. A film may be classified as both action and thriller, or as romance and comedy, requiring models to produce multiple correct predictions per instance rather than a single class label. Early approaches to this problem relied on handcrafted visual features such as color histograms and texture descriptors combined with traditional machine learning classifiers; however, the emergence of deep learning has substantially advanced the state of the art. Convolutional Neural Networks (CNNs), particularly pretrained architectures such as VGG, ResNet, and EfficientNet, have demonstrated strong performance by leveraging transfer learning from large-

scale image datasets [8], [12]. Among these, EfficientNet has emerged as a particularly compelling backbone due to its compound scaling strategy, which simultaneously balances network depth, width, and input resolution to achieve superior accuracy with fewer parameters compared to conventional CNN architectures [12].

Beyond purely visual approaches, researchers have explored multimodal methods that combine poster images with various textual signals to improve genre prediction. These textual modalities range from heavyweight sources such as full plot synopses, subtitles, and cast metadata to lighter signals such as movie titles [5], [6]. The integration of natural language processing techniques, particularly the use of pretrained language models such as Bidirectional Encoder Representations from Transformers (BERT), has enabled richer semantic representations of textual inputs [11]. Nevertheless, the fusion strategy between visual and textual modalities remains an open research question. Simple concatenation of feature vectors is the most widely adopted approach; however, it treats the two modalities as independent and fails to model their mutual interactions, potentially leaving cross-modal dependencies unexploited.

Despite these advances, several research gaps remain. First, most existing studies that incorporate textual modalities rely on heavyweight inputs such as full plot synopses or subtitles, which are not always available at the prerelease stage. The contribution of a lightweight textual signal, specifically the movie title encoded via BERT, remains largely underexplored as a standalone complement to visual features. Second, while cross-attention has proven effective for modeling inter-modal relationships in other vision-language tasks, its application to poster-based genre classification has received limited attention. Third, systematic studies that isolate and quantify the marginal contribution of title information under a controlled experimental setup are scarce, making it difficult for practitioners to assess whether the added complexity of multimodal fusion is justified for this specific task.

This paper addresses the above gaps through a focused comparative study on multilabel movie genre classification from poster images. We adopt EfficientNet-B3 as the visual backbone and investigate two experimental configurations: (1) an image-only baseline that classifies genre exclusively from the poster, and (2) a multimodal model that incorporates the movie title encoded via BERT. In the multimodal configuration, visual and textual features are fused through a cross-attention mechanism, allowing the model to selectively attend to visual regions that are most semantically relevant to the title embedding. Both configurations are evaluated on a consistent dataset of 14 genre categories sourced from the Internet Movie Database (IMDb), enabling a fair and controlled comparison that directly quantifies the added value of title-based textual context.

II. LITERATURE REVIEW AND THEORETICAL BACKGROUND

A. Movie Genre Classification

In the film industry, the concept of genre serves as a fundamental framework that categorizes movies sharing common characteristics, subject matters, and methodologies [8]. With the rapid expansion of digital media and streaming platforms, the demand for automatic movie genre classification has grown significantly among consumers and service providers [6]. Developing robust automated classification systems is highly crucial because it not only simplifies massive content organization, but also enables personalized recommendations, genre-based marketing strategies, and improved overall audience experiences [8]. However, automating this process across visual mediums presents unique difficulties.

Identifying movie genres from poster images is a highly challenging task because genres represent abstract, intangible qualities that are not physically present in any single frame or scene [8]. Unlike standard computer vision tasks that target concrete objects, genre classification relies heavily on subtle visual cues such as color choices, textures, facial expressions, and spatial typography to convey narrative themes [7]. Furthermore, because individual films frequently span across multiple thematic categories, this task cannot be treated as a single label problem [7], [8]. Instead, it is systematically formulated as a multi-label classification problem, requiring computational models to accurately assign and predict one or more genre labels simultaneously based on the visual characteristics of a single poster [7], [8].

Specifically for direct visual feature extraction from movie posters, recent research highlights two highly effective Convolutional Neural Network (CNN) approaches for multi-label genre classification. First, a customized 7-layer CNN architecture, inspired by VGG models and optimized for key hyperparameters, successfully achieved an accuracy of 90.5% on a large-scale dataset of approximately 44,000 posters [7]. Second, a transfer learning approach evaluating modern pre-trained models demonstrated that DenseNet consistently outperforms other architectures, recording an accuracy of up to 92.56% due to its dense connectivity pattern that maximizes feature reuse [8].

B. Multilabel Classification

In machine learning, multi-label classification extends conventional single-label systems by allowing an individual data instance to be simultaneously assigned multiple, non-mutually exclusive class tags. In the context of multi-label classification, selecting an appropriate loss function is critical for ensuring optimal model convergence. While Categorical Cross Entropy is commonly used for mutually exclusive multi-class problems, it is suboptimal for multi-label tasks because it distributes probabilities across classes, effectively forcing them to compete. Instead, Binary Cross Entropy (BCE) is the

standard approach, as it evaluates the probability of each label independently [7]. Mathematically, the BCE loss is defined as:

$$L = -\frac{1}{N} \sum_{i=1}^N [y_i \log(\hat{y}_i) + (1 - y_i) \log(1 - \hat{y}_i)]$$

where N represents the total number of classes (or instances), y_i is the binary ground truth label for the i -th class (0 or 1), and $\hat{y}_i$ denotes the predicted probability output by the model. By treating each class prediction as an independent Bernoulli distribution, BCE allows for a fair and parallel computation of loss across all labels. This makes it highly effective for capturing complex visual semantics, such as movie posters, where a single input often encapsulates multiple thematic genres simultaneously.

C. EfficientNet and Compound Scaling

TABLE I
EFFICIENTNET-B0 BASELINE NETWORK. EACH ROW DESCRIBES STAGE i WITH $\hat{L}_i$ LAYERS, INPUT RESOLUTION $(\hat{H}_i, \hat{W}_i)$, AND OUTPUT CHANNELS $\hat{C}_i$.

Stage i	Operator $\hat{F}_i$	Resolution $\hat{H}_i \times \hat{W}_i$	#Channels $\hat{C}_i$	#Layers $\hat{L}_i$
1	Conv 3×3	224×224	32	1
2	MBConv1, $k3 \times 3$	112×112	16	1
3	MBConv6, $k3 \times 3$	112×112	24	2
4	MBConv6, $k5 \times 5$	56×56	40	2
5	MBConv6, $k3 \times 3$	28×28	80	3
6	MBConv6, $k5 \times 5$	14×14	112	3
7	MBConv6, $k5 \times 5$	14×14	192	4
8	MBConv6, $k3 \times 3$	7×7	320	1
9	Conv 1×1 + Pooling + FC	7×7	1280	1

Conventional approaches to scaling Convolutional Neural Networks (ConvNets) typically focus on expanding a single dimension such as depth, width, or image resolution which often requires tedious manual tuning and leads to suboptimal accuracy and efficiency. To address this limitation, researchers introduced a novel "compound scaling" method that systematically balances and uniformly scales network depth, width, and resolution using a constant, principled coefficient. Utilizing this compound scaling method alongside Neural Architecture Search (NAS) to optimize both predictive accuracy and computational operations (FLOPS), a highly efficient baseline architecture named EfficientNet was developed. The scaled-up EfficientNet models demonstrated exceptional performance across tasks notably, EfficientNet-B7 achieved a state-of-the-art 84.3% top-1 accuracy on ImageNet while being 8.4 times smaller and 6.1 times faster during inference compared to previous top models like GPipe [12].

D. Attention Mechanism

The Transformer architecture replaces traditional recurrent and convolutional layers entirely with attention mechanisms to capture global dependencies [9]. Its core function maps queries

and key-value pairs to a weighted output, operationalized mathematically as Scaled Dot-Product Attention:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V \quad (1)$$

Scaling by $\sqrt{d_k}$ is critical to maintain stable gradients during softmax computation [9]. Furthermore, the model employs Multi-Head Attention, which linearly projects inputs into multiple learned subspaces to process attention in parallel, enabling it to jointly attend to diverse information from different representational spaces simultaneously [9].

E. BERT for Text Encoding

BERT (Bidirectional Encoder Representations from Transformers) establishes a new standard for contextual language representation by pre-training deep bidirectional representations from unlabeled text. Unlike previous unidirectional models that only condition on prior tokens, BERT utilizes a Masked Language Model (MLM) objective, which allows the architecture to seamlessly fuse both left and right contexts across all layers [11]. This deep bidirectionality enables the model to capture complex semantic relationships and nuanced contextual meanings within a sentence.

A major strength of BERT lies in its exceptional capability to generate highly contextualized word embeddings through a feature-based approach. Instead of fine-tuning the entire architecture, fixed contextual representations can be extracted directly from the pre-trained model. Empirical evaluations demonstrate that extracting and concatenating token representations from the top four hidden layers produces rich, robust embeddings that achieve highly competitive performance on downstream tasks (such as Named Entity Recognition). This proves BERT's remarkable flexibility not just as an end-to-end model, but as a powerful feature extractor for generating contextual word embeddings.

F. Test-Time Augmentation

Test-Time Augmentation (TTA) is a widely adopted technique in image classification to enhance model robustness during the inference phase by aggregating predictions across multiple transformed versions of a single test input [10]. Traditionally, these predictions from various augmented variants such as subtle rotations, crops, or color adjustments are combined using a simple average mechanism [10]. Although standard TTA consistently yields a net improvement in overall classification accuracy, experimental analyses reveal that a simple average aggregation is frequently suboptimal and can inadvertently flip several initially correct predictions into incorrect ones [10]. Consequently, recent advancements focus on developing learning-based aggregation methods to integrate augmented predictions more effectively, minimizing misclassifications and providing more stable, generalized outputs for complex multi-label visual tasks [10].

III. DATASET AND EXPLORATORY DATA ANALYSIS

A. Dataset Description

The dataset used in this study consists of high-resolution RGB images of movie posters sourced from the Internet Movie Database (IMDb). Each poster is densely packed with diverse visual elements, including human faces, objects, typography, and genre-specific color grading, making them rich inputs for visual genre inference. The dataset covers 14 genre categories: *action*, *adventure*, *animation*, *comedy*, *crime*, *drama*, *family*, *fantasy*, *horror*, *musical*, *mystery*, *romance*, *sci-fi*, and *thriller*.

The full dataset contains **300 labeled movie poster images**, split into **250 images (83%)** for training and **50 images (17%)** for the held-out test set. Because each film may belong to multiple genres simultaneously, this is a multi-label classification task; genre labels are non-mutually exclusive binary vectors of length 14. The training set is further partitioned into train and validation subsets using `MultilabelStratifiedShuffleSplit` (80/20 ratio, random state 42) to preserve label distribution across all splits.

B. Genre Distribution

Figure 1 shows the count of positive samples per genre in the training set. The distribution is **highly imbalanced**, with a mean of approximately 69 samples per genre. *Action* (167), *drama* (146), and *adventure* (125) dominate the dataset, each appearing in more than 50% of all posters. At the opposite extreme, *horror* (22) and *musical* (5) are severely underrepresented with fewer than 30 positive samples, placing them in the minority class regime.

This pronounced imbalance has direct implications for model training. Dominant genres can bias the classifier toward high recall on frequent labels while suppressing minority-class predictions. To address this, per-class positive weights (Eq. 2) and targeted oversampling of minority samples are applied during training, as described in Section IV.

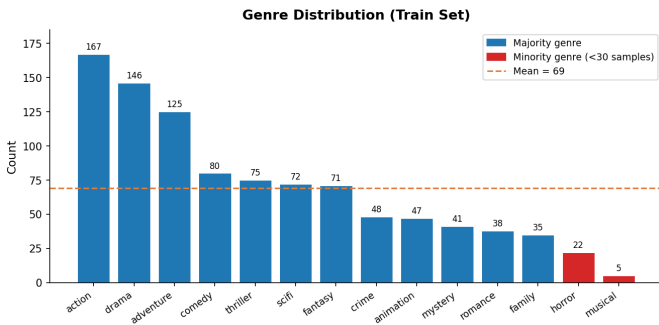


Fig. 1. Genre distribution in the training set. Red bars indicate minority genres with fewer than 30 positive samples. The dashed line marks the mean count of 69.

C. Label Co-occurrence

Figure 2 presents the normalized genre co-occurrence matrix computed over the training set. Each cell (i, j) represents

the fraction of films labeled with genre i that also carry genre j .

Several strong co-occurrence patterns emerge. *Sci-fi* films are overwhelmingly labeled as *action* (0.93) and *adventure* (0.81), reflecting the typical narrative structure of blockbuster science fiction. *Animation* films co-occur heavily with *adventure* (0.94) and *family* (0.60), consistent with the family-friendly nature of animated features. *Crime* films frequently co-occur with *drama* (0.73) and *thriller* (0.67), while *thriller* strongly correlates with *drama* (0.72) and *action* (0.69).

These high inter-label correlations indicate that movie genres are **not independent**, and a classifier that exploits label dependencies may achieve higher accuracy than one that treats each label in isolation. While the current architecture uses independent sigmoid outputs per label, the co-occurrence patterns also provide a useful diagnostic for interpreting per-genre errors.

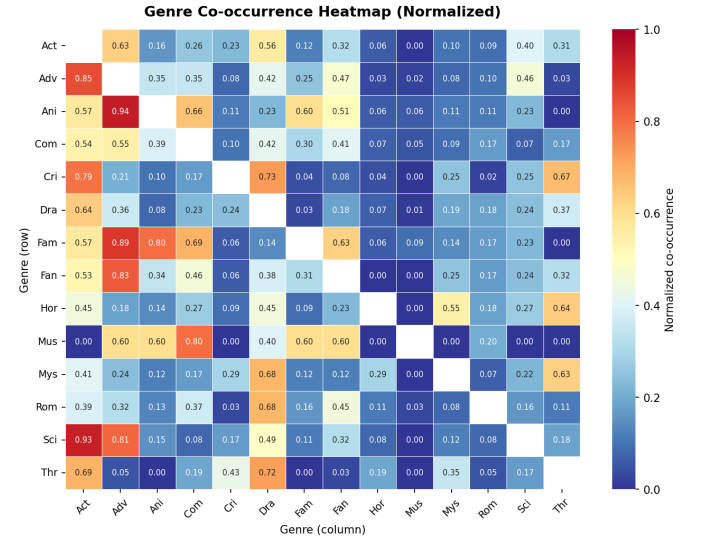


Fig. 2. Normalized genre co-occurrence heatmap over the training set. Cell (i, j) gives the fraction of genre- i films that also belong to genre j . Abbreviations: Act=action, Adv=adventure, Ani=animation, Com=comedy, Cri=crime, Dra=drama, Fam=family, Fan=fantasy, Hor=horror, Mus=musical, Mys=mystery, Rom=romance, Sci=sci-fi, Thr=thriller.

D. Distribution of Genre Count per Film

Figure 3 shows the distribution of the number of genre labels assigned to each film in the training and test sets. Both splits exhibit nearly identical distributions, with a mean of 3.89 (train) and 3.86 (test) genres per film, confirming that the stratified split successfully preserves label-count statistics.

Almost no film in the dataset carries a single genre label; the vast majority of samples have between 3 and 5 co-occurring genres, with a peak at 4 labels (90 training samples, 23 test samples). This characteristic severely rules out single-label formulations and underscores the necessity of a multi-label approach with per-label binary cross-entropy loss.

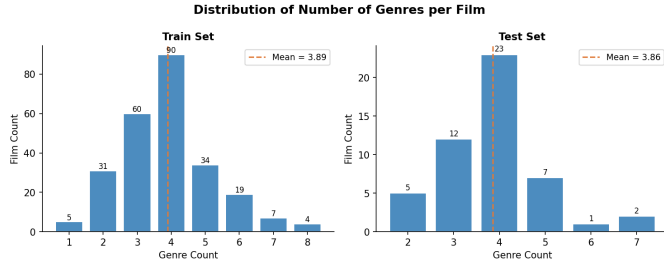


Fig. 3. Distribution of genre count per film for the training set (left) and test set (right). The dashed line marks the mean. Both splits peak at 4 genres per film, confirming consistent label-count distribution.

E. Visual Feature Analysis: Brightness by Genre

To examine whether low-level visual statistics differ meaningfully across genres, Figure 4 presents the distribution of mean poster brightness (pixel intensity averaged over RGB channels, range 0–255) per genre.

A clear separation is visible at the extremes. *Musical* posters are noticeably brighter overall, exhibiting a higher median brightness and a tighter, more elevated interquartile range relative to other genres. This is consistent with the typically colorful and vibrant visual language associated with musical productions. In contrast, *horror*, *crime*, and *mystery* posters tend to be visually darker, with lower median brightness scores, reflecting their conventional use of dark color palettes and moody lighting to convey tension and unease.

These observations corroborate the hypothesis that genre-specific visual conventions are encoded in low-level poster statistics, motivating the use of a deep convolutional backbone capable of capturing both low-level (color, texture) and high-level (composition, semantic) visual features.

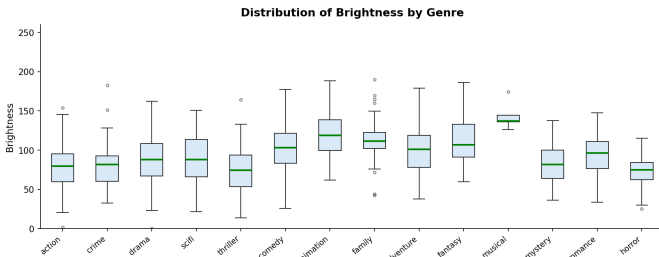


Fig. 4. Distribution of mean poster brightness per genre. Musicals display the highest median brightness, while horror, crime, and mystery genres are consistently darker, reflecting their genre-specific visual conventions.

F. Class Imbalance Analysis and Mitigation

Table II summarizes the positive sample counts per genre across all splits. Three genres, *musical* (5 train), *horror* (22 train), and *romance* (38 train) — fall below or near the minority threshold of 30 samples. Given the 80/20 train-validation split, the effective positive count available for gradient updates is even smaller after stratification.

To partially mitigate the imbalance, two complementary strategies are employed. First, samples containing any genre

TABLE II
POSITIVE SAMPLE COUNT PER GENRE (TRAINING SET)

Genre	Train Count	% of Total Films
Action	167	66.8%
Drama	146	58.4%
Adventure	125	50.0%
Comedy	80	32.0%
Thriller	75	30.0%
Sci-fi	72	28.8%
Fantasy	71	28.4%
Crime	48	19.2%
Animation	47	18.8%
Mystery	41	16.4%
Romance	38	15.2%
Family	35	14.0%
Horror	22	8.8%
Musical	5	2.0%
Mean	69	27.6%

TABLE III
IMAGE AUGMENTATION PIPELINE (TRAINING ONLY)

Transform	Parameter	Purpose
RandomResizedCrop	scale=(0.85,1.0)	Crop variation
RandomRotation	degrees=5	Light rotation
ColorJitter	brightness=0.10	Color variation
GaussianBlur	kernel=3, p=0.2	Blur simulation
RandHorizontalFlip	p=0.5	Symmetry flip
Normalize	ImageNet stats	Normalization

with fewer than 15 occurrences are duplicated five times in the training set (targeted oversampling), ensuring that minority-class gradients are amplified relative to the default data distribution. Second, per-class positive weights proportional to the negative-to-positive ratio (Eq. 2) are applied to the BCEWithLogitsLoss, penalizing false negatives on underrepresented genres more heavily than false negatives on majority genres. Despite these measures, the extremely low positive count of *musical* (5 training samples) renders reliable classification practically infeasible, as confirmed by the zero F1-score on this genre in both experimental configurations (see Section V).

IV. METHODOLOGY

A. Data Preprocessing

1) *Image Preprocessing*: All poster images are resized to 300×300 pixels and normalized using ImageNet statistics ($\mu = [0.485, 0.456, 0.406]$, $\sigma = [0.229, 0.224, 0.225]$). Data augmentation is applied exclusively to the training split, as summarized in Table III.

2) *Text Preprocessing (Experiment I Only)*: Movie titles are tokenized using BertTokenizer from bert-base-uncased with max_length=16, padding, and truncation enabled. The tokenizer produces input_ids and attention_mask as inputs to BERT.

TABLE IV
MODEL COMPONENTS, EXPERIMENT I (MULTIMODAL)

Component	Detail	Dim
bert	BertModel, last 4 CLS avg	768
eff_features	EfficientNetB3 conv	1536
eff_avgpool	trunk AdaptiveAvgPool2d(1)	1536
img_proj	Linear(1536, 768)	768
cross_attn	MultiheadAttn(768, heads=8)	768
layer_norm	LayerNorm(768)	768
classifier[0]	Linear(2304, 1024)	1024
classifier[1-3]	LayerNorm, ReLU, Dropout(0.3)	1024
classifier[4]	Linear(1024, 14)	14
Total parameters (est.)		$\pm 124M$

TABLE V
MODEL COMPONENTS, EXPERIMENT II (IMAGE-ONLY)

Component	Detail	Dim
backbone	EfficientNetB3 + Avg- Pool	1536
classifier[0]	Linear(1536, 512)	512
classifier[1-3]	LayerNorm, ReLU, Dropout(0.3)	512
classifier[4]	Linear(512, 14)	14
Total parameters (est.)		$\pm 10.7M$

B. Model Architecture

Two experimental configurations are evaluated, both using EfficientNet-B3 as the visual backbone.

1) *Experiment I: Multimodal EfficientNet-B3 + BERT + Cross-Attention*: The multimodal model processes two parallel streams. The visual stream passes the poster through EfficientNet-B3 (ImageNet pretrained), producing a 1536-d feature via AdaptiveAvgPool2d(1), projected to 768-d via a linear layer. The textual stream encodes the movie title through BERT-base-uncased, averaging the last four layers' [CLS] tokens to produce a 768-d embedding.

The two embeddings are fused via an 8-head cross-attention mechanism, where the text embedding is the query and the projected image embedding provides keys and values. The cross-attended output is layer-normalized and concatenated with the original text and image embeddings, yielding a $768 + 768 + 768 = 2304$ -d fused vector fed into the classifier head. The full component breakdown is shown in Table IV.

2) *Experiment II: Image-Only EfficientNet-B3*: The image-only baseline passes the 1536-d EfficientNet-B3 feature directly to a two-layer classifier without any text input or fusion, as detailed in Table V. This configuration is kept intentionally compact so that any performance difference can be attributed to the added title-aware fusion path rather than to an uncontrolled change in the visual backbone.

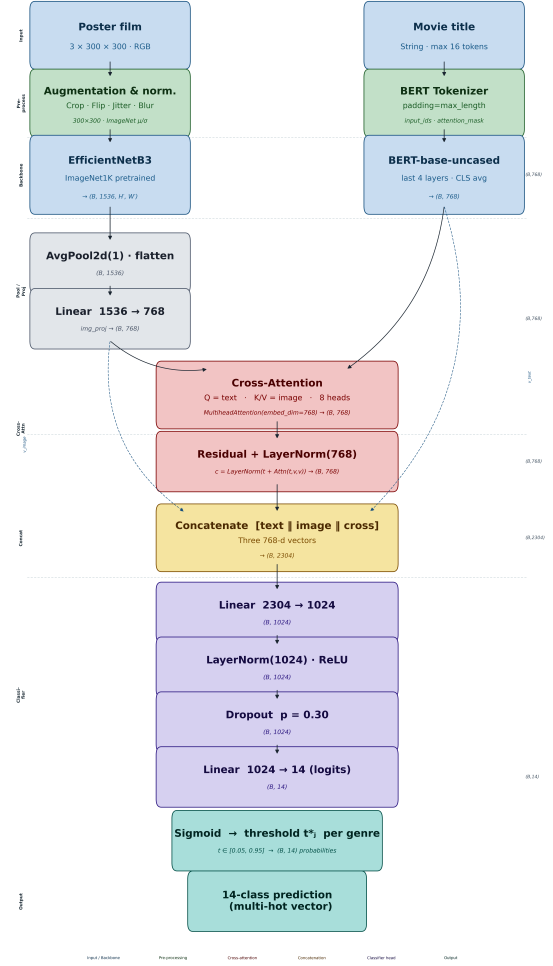


Fig. 5. Architecture of Experiment I: EfficientNet-B3 + BERT + Cross-Attention Fusion.

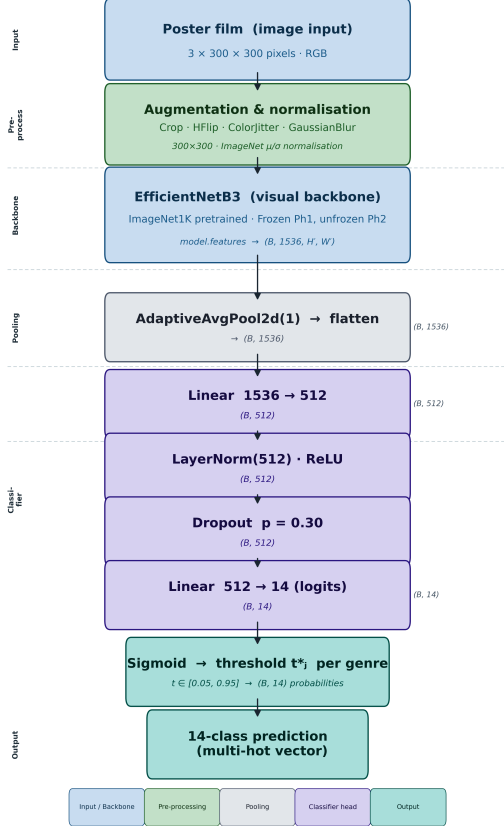


Fig. 6. Architecture of Experiment II: EfficientNet-B3 Image-Only.

C. Experiment Setup

1) Data Strategy:

a) *Train/Validation/Test Split.*: The dataset is split using MultilabelStratifiedShuffleSplit (80% train / 20% validation, random state 42), preserving multilabel distribution across splits. The held-out test set is used solely for final evaluation.

b) *Targeted Oversampling.*: Samples containing any genre with fewer than 15 occurrences are duplicated $\times 5$ on the training set only, preventing data leakage into validation and test sets.

c) *Positive Weight.*: Per-class positive weights are applied to the loss to penalize false negatives on minority genres:

$$w_i = \frac{N - n_i^+}{n_i^+ + \varepsilon}, \quad \varepsilon = 10^{-5} \quad (2)$$

where N is the total training samples and n_i^+ is the positive sample count for genre i .

2) *Loss Function.*: BCEWithLogitsLoss with per-class positive weights is used as the primary loss function:

$$\mathcal{L} = -\frac{1}{N \cdot C} \sum_{i=1}^N \sum_{j=1}^C [w_j y_{ij} \log \sigma(z_{ij}) + (1 - y_{ij}) \log (1 - \sigma(z_{ij}))] \quad (3)$$

TABLE VI
COMPLETE HYPERPARAMETER CONFIGURATION

Parameter	Value	Exp.
val_ratio	0.2	I, II
minority_threshold	15	I, II
oversample_factor	5	I, II
batch_size	16	I, II
image_size	300	I, II
bert_name	bert-base-uncased	I
max_text_len	16	I
epochs (Ph.1 / Ph.2)	30 / 60	I, II
lr_phase1	5×10^{-4}	I, II
lr_backbone	10^{-5}	I, II
lr_classifier	5×10^{-5}	I, II
weight_decay	0.01	I, II
dropout	0.3	I, II
Scheduler Ph.1	CosineAnnealingLR	I, II
Scheduler Ph.2	OneCycleLR	I, II

where $y_{ij} \in \{0, 1\}$ is the ground-truth label, z_{ij} is the raw logit, and $\sigma(\cdot)$ is the sigmoid function. Asymmetric Loss (ASL) is also implemented as an alternative.

3) *Two-Phase Training*: Training is conducted in two phases to prevent catastrophic forgetting.

Phase 1 — Backbone Frozen (30 epochs): The backbone is frozen in evaluation mode. Only the head (and fusion modules in Experiment I) are trained using CosineAnnealingLR:

$$\eta(t) = \eta_{\min} + \frac{1}{2}(\eta_0 - \eta_{\min}) \left(1 + \cos \frac{\pi t}{T_{\max}}\right) \quad (4)$$

with $\eta_0 = 5 \times 10^{-4}$, $\eta_{\min} = 10^{-6}$, $T_{\max} = 30$.

Phase 2 — Full Fine-Tuning (60 epochs): All parameters are unfrozen from the best Phase 1 checkpoint. Differential learning rates are applied: backbone at 10^{-5} and head/fusion at 5×10^{-5} (ratio of 5 $\times$). OneCycleLR is used per step with 15% warm-up and cosine decay.

The complete hyperparameter configuration is listed in Table VI.

4) *Threshold Tuning Per-Genre*: The decision threshold is independently optimized per genre on the validation set by sweeping $t \in [0.05, 0.95]$ with step 0.01 and selecting $t_j^* = \arg \max_t F1(y_j, \hat{y}_j(t))$. The optimized thresholds are then applied during test set evaluation.

5) *Test-Time Augmentation*: To improve prediction robustness, $K=5$ augmented views per image are averaged at inference:

$$\bar{p}_j = \frac{1}{K} \sum_{k=1}^K p_j^{(k)}, \quad K = 5 \quad (5)$$

D. Evaluation Metrics

a) Macro F1-Score (primary).:

$$F1_{\text{macro}} = \frac{1}{C} \sum_{j=1}^C F1_j \quad (6)$$

Treats all genres equally regardless of frequency, making it suitable for imbalanced multilabel evaluation.

b) *Micro F1-Score (secondary)*.: Aggregates TP, FP, FN across all genres before computing F1, giving more weight to frequent genres.

c) *Hamming Loss*.:

$$\mathcal{L}_{\text{Hamming}} = \frac{1}{N \cdot C} \sum_{i=1}^N \sum_{j=1}^C \mathbf{1}[\hat{y}_{ij} \neq y_{ij}] \quad (7)$$

d) *AUC-ROC*.: Macro-averaged area under the ROC curve, computed per genre to assess ranking performance independent of threshold.

V. RESULTS AND DISCUSSION

A. Quantitative Results

Table VII presents the overall performance of both experimental configurations on the held-out test set. For each model, two evaluation scenarios are reported: a uniform threshold of $t = 0.5$ applied identically to all genres, and per-genre optimized thresholds obtained through F1-score maximization on the validation set ($t \in [0.05, 0.95]$, step 0.01).

The results reveal a substantial and consistent performance gap between the two architectures. At the default threshold of $t = 0.5$, the multimodal model (Experiment I) achieves a Macro F1 of 0.6502, compared to only 0.4554 for the image-only baseline (Experiment II), a margin of +0.1948. This gap persists after per-genre threshold optimization, with Experiment I reaching a Macro F1 of 0.7293 against 0.5269 for Experiment II, confirming that the multimodal advantage is robust and not an artifact of threshold calibration.

Beyond Macro F1, the multimodal model demonstrates consistently superior behavior across all reported metrics at the default threshold. It achieves a Micro F1 of 0.8385 versus 0.5598 for the image-only model, a Weighted F1 of 0.8222 versus 0.6381, and a dramatically lower Hamming Loss of 0.0814 compared to 0.3257. The Exact Match Ratio of 56.00% for Experiment I versus 0.00% for Experiment II is particularly striking: the image-only baseline fails to produce a single fully correct multilabel prediction on the test set at the default threshold, suggesting that without title context, the model’s per-genre confidence scores are poorly calibrated for the joint multilabel decision boundary.

After threshold optimization, the image-only model achieves an Exact Match Ratio of 0.00% and a Hamming Loss of 0.3400, both of which are worse than the multimodal model at the same setting (0.30, 0.0957). This highlights that threshold tuning yields a selective gain for Experiment II, improving Macro F1 by +0.0715 by recovering minority-class predictions, but cannot overcome the underlying weakness in raw feature discrimination.

Notably, threshold optimization provides a larger relative gain for Experiment II (+0.0715 Macro F1) than for Experiment I (+0.0791), but the absolute scores remain in favor of the multimodal model across all thresholds. The consistent superiority of Experiment I across all metrics strongly supports the hypothesis that title-based textual context provides complementary and discriminative information that poster visual features alone cannot replicate.

TABLE VII
OVERALL PERFORMANCE ON TEST SET

Model	Thr.	Macro F1	Micro F1	Weighted F1	Hamming	Exact Match
Exp. I (Multimodal)	0.5	0.6502	0.8385	0.8222	0.0814	56.00%
Exp. I (Multimodal)	optimal	0.7293	0.8393	0.8837	0.0957	30.00%
Exp. II (Image-Only)	0.5	0.4554	0.5598	0.6381	0.3257	0.00%
Exp. II (Image-Only)	optimal	0.5269	0.5810	0.7055	0.3400	0.00%

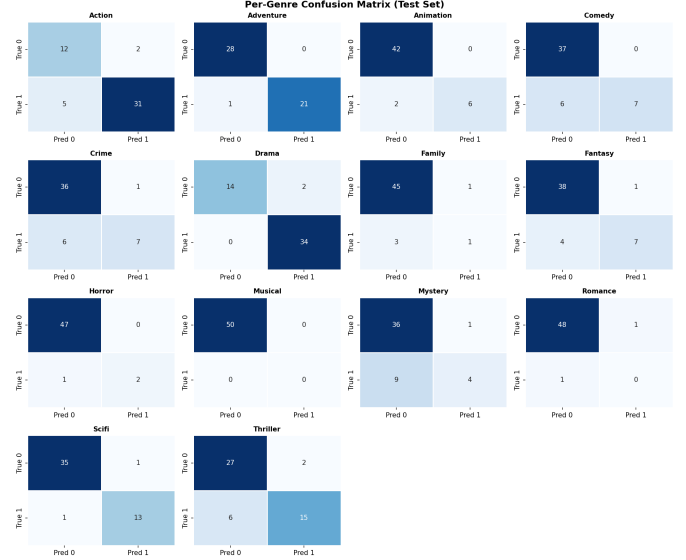


Fig. 7. Per-genre confusion matrices, Experiment I (Multimodal). Each 2×2 matrix shows TN/FP/FN/TP counts on the test set at the per-genre optimal threshold.

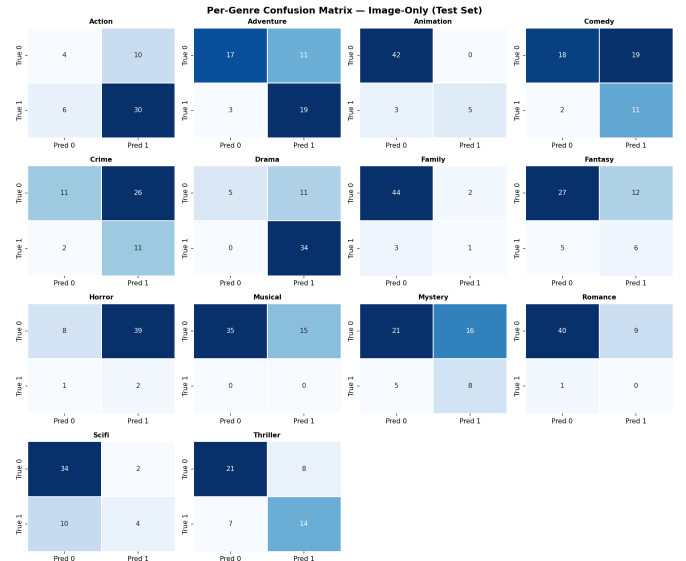


Fig. 8. Per-genre confusion matrices, Experiment II (Image-Only).

B. Per-Genre Analysis

Table VIII reports the per-genre F1 scores for both models at the default ($t = 0.5$) and per-genre optimal thresholds, alongside the optimal threshold t_j^* selected for each genre on the validation set.

TABLE VIII
PER-GENRE F1 COMPARISON BEFORE AND AFTER THRESHOLD OPTIMIZATION FOR EXPERIMENT I (MULTIMODAL) AND EXPERIMENT II (IMAGE-ONLY)

Genre	Exp. I, Multimodal			Exp. II, Image-Only		
	t^*	F1 @ 0.50	F1 @ Optimal	t^*	F1 @ 0.50	F1 @ Optimal
Action	0.15	0.8986	0.9351	0.05	0.7895	0.8372
Adventure	0.38	0.9767	0.9778	0.44	0.7308	0.7556
Animation	0.29	0.8571	0.9333	0.60	0.7692	0.8571
Comedy	0.47	0.7000	0.9231	0.39	0.5116	0.6452
Crime	0.13	0.6667	0.7857	0.32	0.4400	0.4681
Drama	0.39	0.9714	0.9855	0.45	0.8608	0.8684
Family	0.89	0.3333	0.4000	0.81	0.2857	0.3333
Fantasy	0.76	0.7368	0.7619	0.37	0.4138	0.5000
Horror	0.78	0.8000	1.0000	0.40	0.0909	0.1429
Musical	0.05	0.0000	0.0000	0.05	0.0000	0.0000
Mystery	0.10	0.4444	0.6207	0.36	0.4324	0.4898
Romance	0.06	0.0000	0.0870	0.23	0.0000	0.0556
Sci-fi	0.68	0.9286	0.9630	0.60	0.4000	0.7200
Thriller	0.31	0.7895	0.8372	0.45	0.6512	0.7037
Macro F1	0.39	0.6502	0.7293	0.39	0.4554	0.5269

Note: t^* denotes the optimal decision threshold selected independently for each genre based on validation F1-score maximization.

Several observations emerge from the per-genre analysis.

1) *Genres with Zero or Near-Zero Test Positives*: Both models completely fail on *musical* ($F1 = 0.00$) and *romance* ($F1 \leq 0.09$) regardless of architecture or threshold strategy. Inspection of the confusion matrices confirms this: for *musical*, Experiment I (Fig. 7) shows 50 TN and 0 TP, with all test instances falling into the negative class, meaning there are zero positive test samples for this genre. For *romance*, only 1 positive test sample exists, and it is misclassified as a negative by both models.

These failures are a direct consequence of extreme class sparsity. With only 5 training samples for *musical* and 38 for *romance*, neither targeted oversampling nor positive-weight BCE can compensate for the near-absence of a meaningful training signal. This is a dataset-level limitation that no architectural improvement can resolve.

2) *Title-Sensitive Genres*: The most pronounced gains for the multimodal model are observed on genres that carry strong and consistent lexical signals in their titles. *Sci-fi* shows a dramatic improvement: at the default threshold, Experiment I achieves $F1 = 0.9286$ versus only 0.4000 for Experiment II, with further improvement to 0.9630 versus 0.7200 after optimization. Similarly, *horror* achieves $F1 = 0.8000$ versus 0.0909 (at $t = 0.5$) and 1.0000 versus 0.1429 after optimization, a near-perfect score for the multimodal model versus near-complete failure for the image-only baseline.

Inspection of the confusion matrices supports these figures. For *horror*, Experiment I (Fig. 7) records 47 TN, 0 FP, 1 FN, and 2 TP at the optimal threshold, indicating strong true-negative specificity with minimal false positives, while Experiment II (Fig. 8) records 8 TN, 39 FP, 1 FN, and 2 TP, a massive false-positive rate that severely degrades precision. This pattern is consistent with the hypothesis that horror

posters may visually overlap with other dark-toned genres (crime, thriller, mystery), but horror-specific title words provide the disambiguation necessary for accurate classification.

Adventure and *fantasy* show similar patterns. *Adventure* reaches $F1 = 0.9778$ in Experiment I versus 0.7556 in Experiment II after optimization, while *fantasy* achieves 0.7619 versus 0.5000. For *sci-fi*, the confusion matrix of Experiment II (Fig. 8) reveals 34 TN but 2 FP, 10 FN, and only 4 TP at the optimal threshold, demonstrating that the image-only model fails to identify the majority of sci-fi positives, whereas Experiment I recovers 13 TP with only 1 FN and 1 FP (Fig. 7).

These observations are consistent with the findings of the GradCAM analysis (Section V-D), which demonstrates that the cross-attention mechanism guides EfficientNet-B3 to attend to world-building background elements, spacecraft, cosmic environments, supernatural settings, that are highly informative for these genres when cued by the title embedding.

3) *Visually Dominant Genres*: For genres where visual cues are strong and consistent across posters, both models perform comparably, and the multimodal advantage narrows. *Drama* achieves $F1 = 0.9855$ (Experiment I) versus 0.8684 (Experiment II) after optimization, both high, with the gap attributable to the multimodal model’s more precise threshold selection ($t^* = 0.39$ vs. 0.45). Confusion matrices confirm that both models produce near-zero false negatives on drama (Fig. 7 and Fig. 8), reflecting the genre’s high frequency and visually diverse but still distinctive poster characteristics (e.g., portrait compositions, muted palettes).

Animation also shows comparable strong performance for both models ($F1 = 0.9333$ vs. 0.8571 after optimization), as animation posters are visually distinctive with stylized character designs and vivid color palettes that serve as robust visual classifiers independently of title information.

Crime and *mystery*, while semantically related, display a moderate multimodal advantage (0.7857 vs. 0.4681 and 0.6207 vs. 0.4898 respectively). The confusion matrix for Experiment II shows a high false-positive rate for crime (Fig. 8: 26 FP versus 1 FP in Experiment I), likely because crime poster aesthetics (dark tones, urban settings, characters in formal attire) visually overlap with drama and thriller.

4) *Genres with Low Overall Performance*: *Family* remains a weak genre for both models despite its moderate training count (35 samples). F1 scores of 0.4000 (Experiment I) and 0.3333 (Experiment II) after optimization reflect its high visual variability: family-genre posters may feature animated characters, live-action children’s adventures, or broad comedies, making them visually indistinct as a genre cluster. Both confusion matrices show very few true positives (1 each), indicating that the models overwhelmingly default to predicting the negative class.

Taken together, the per-genre results indicate that the proposed multimodal architecture is most beneficial when title semantics can resolve visual ambiguity, while gains are smaller for categories that are either already visually distinctive or too underrepresented to provide reliable learning signals. This interpretation motivates the training-dynamics analysis in the next subsection, which examines whether the observed performance differences are supported by the optimization behavior of both models.

C. Training Dynamics

1) *Phase 1, Frozen Backbone*: Fig. 9 and Fig. 10 show the Phase 1 training and validation loss curves for Experiments I and II, respectively. Both models exhibit a characteristic divergence pattern: training loss decreases sharply, converging to near zero within the first 10 epochs, while validation loss increases monotonically throughout all 30 epochs, rising from approximately 0.85 to 1.20 (Experiment I) and 1.15 (Experiment II).

This behavior is a direct consequence of the frozen-backbone constraint: only the classifier head (and cross-attention fusion module in Experiment I) is optimized, while EfficientNet-B3 and BERT weights remain fixed. With the backbone unable to adapt its representations to the dataset distribution, the trainable classifier rapidly memorizes augmented training samples, a process accelerated by the positive-weight BCE, which strongly penalizes false negatives on minority classes and incentivizes high-confidence positive predictions.

The divergence is more pronounced in Experiment I than Experiment II. The multimodal model’s larger trainable head (encompassing the 8-head cross-attention, layer normalization, and a $2304 \rightarrow 1024 \rightarrow 14$ classifier) provides greater capacity for overfitting relative to the simpler $1536 \rightarrow 512 \rightarrow 14$ head of the image-only model. Nevertheless, the Phase 1 checkpoint serves its intended purpose: it initializes the head with reasonable feature-to-label mappings before the higher-risk full fine-tuning in Phase 2.

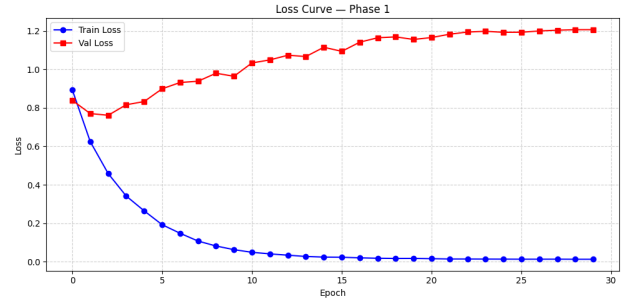


Fig. 9. Loss curve, Phase 1, Experiment I (Multimodal).

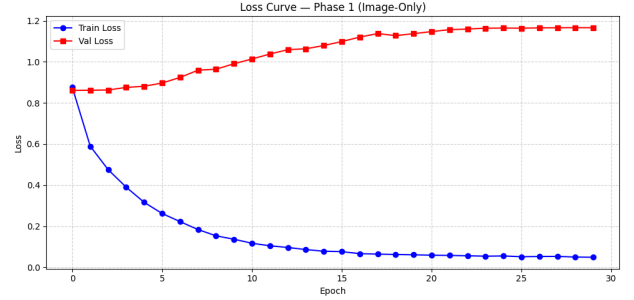


Fig. 10. Loss curve, Phase 1, Experiment II (Image-Only).

2) *Phase 2, Full Fine-Tuning*: The Phase 2 dynamics differ markedly between the two experiments, as shown in Fig. 11 and Fig. 12.

For Experiment II (image-only, Fig. 12), Phase 2 exhibits a healthy and stable fine-tuning pattern. Training loss decreases gradually from approximately 0.85 to 0.34 over 60 epochs, following the smooth trajectory imposed by OneCycleLR. Crucially, validation loss also decreases initially, dropping from ≈ 0.90 to ≈ 0.84 within the first 5 epochs, before stabilizing around 0.86 through the remainder of training. The absence of a significant train-validation gap after epoch 10 indicates that EfficientNet-B3 fine-tuning successfully adapts the backbone’s visual representations to the poster domain while maintaining generalization. The differential learning rate strategy (backbone at 10^{-5} , head at 5×10^{-5}) appears effective in preventing catastrophic forgetting of ImageNet features while enabling task-specific specialization.

For Experiment I (multimodal, Fig. 11), Phase 2 reveals a qualitatively different dynamic. Training loss decreases steeply from ≈ 0.57 to ≈ 0.07 over 60 epochs, a substantially lower final training loss than Experiment II, while validation loss follows a trajectory with an initial drop from ≈ 0.74 to ≈ 0.53 within the first 15 epochs, followed by a plateau and stabilization around 0.53. Although the model does not exhibit catastrophic divergence, the persistent and widening train-validation gap indicates that the larger parameter space introduced by BERT ($\sim 110M$ parameters) creates an overfitting pressure that the dataset of 250 training samples cannot fully regularize. The weight decay of 0.01 and dropout of 0.3 partially mitigate this, but they are insufficient given the extreme model-to-data ratio.

This observation is consistent with the quantitative result that threshold optimization yields a smaller macro-level gain for Experiment I: the model’s sigmoid outputs for uncertain genres reflect overconfident but poorly calibrated predictions, which limits the benefit of per-genre threshold sweeping.

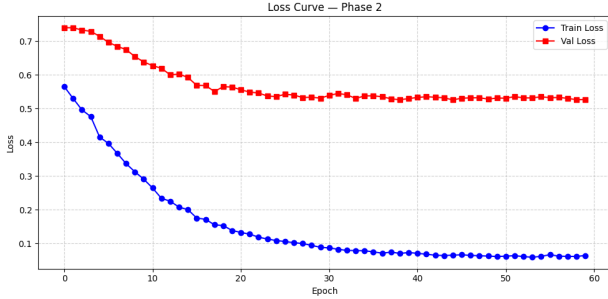


Fig. 11. Loss curve, Phase 2, Experiment I (Multimodal).

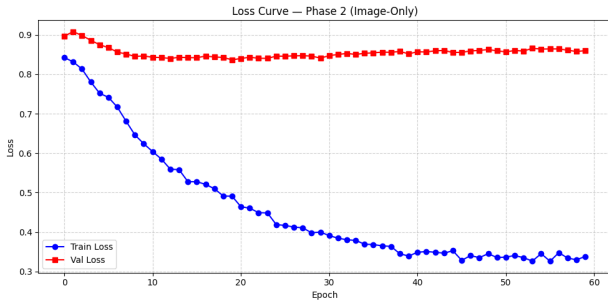


Fig. 12. Loss curve, Phase 2, Experiment II (Image-Only).

D. Visual Explanation: GradCAM Analysis

To qualitatively assess which spatial regions of the poster drive genre predictions, Gradient-weighted Class Activation Mapping (GradCAM) [13] is applied to the final convolutional block of EfficientNet-B3. Fig. 13 presents a representative example from Experiment I using the poster of *Guardians of the Galaxy* (true labels: action, adventure, comedy, sci-fi).

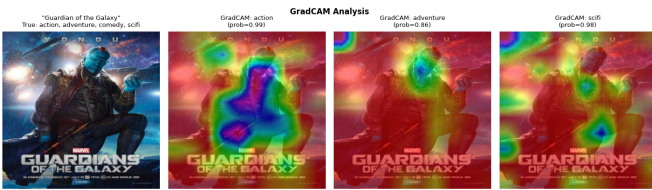


Fig. 13. GradCAM visualizations for *Guardians of the Galaxy* (Experiment I, Multimodal). From left: original poster; activation map for *action* (prob = 0.99), *adventure* (prob = 0.86), and *sci-fi* (prob = 0.98). Warmer colors indicate regions of higher gradient activation.

The GradCAM maps reveal that the model attends to semantically meaningful and genre-discriminative spatial regions, with distinct activation patterns for each predicted genre:

- For **action** (prob = 0.99), the activation map shows intense concentration on the central humanoid figure,

particularly the torso, arms, and weapon held in a combat-ready pose. Secondary activations appear at energy effects and dynamic foreground elements. This is consistent with action genre conventions, where powerful, kinetic character poses serve as the primary visual indicator. The model correctly ignores the background, which is shared with other genres.

- For **adventure** (prob = 0.86), the activation spreads more broadly across the figure and upper background, suggesting the model integrates multiple spatial cues, character silhouette, spacecraft visible in the upper left, and the expansive cosmic environment, rather than attending to a single focal point. This broader integration reflects the compositional nature of adventure genre posters, which typically communicate scope and journey through environmental context alongside character.
- For **sci-fi** (prob = 0.98), the activation pattern shifts notably toward the title region and the spacecraft and cosmic background in the upper portion of the poster. The reduced character-centered activation and increased attention to world-building background elements indicates that sci-fi classification relies on contextual cues such as futuristic technology and deep-space environments rather than character pose alone. This shift is likely mediated by the cross-attention mechanism: the title embedding for “Guardians of the Galaxy” contains strong sci-fi lexical cues (“Galaxy”) that guide the visual attention toward cosmologically informative regions.

These attention patterns demonstrate that EfficientNet-B3’s final convolutional features learn spatially structured, genre-discriminative representations. Importantly, the activation differences between genres on the same poster confirm that the model is not learning a single global feature vector, but rather genre-specific spatial patterns, a property that the cross-attention mechanism in Experiment I can selectively amplify based on the title embedding. The qualitative evidence from GradCAM thus provides mechanistic support for the quantitative advantage of the multimodal architecture on lexically distinctive genres such as sci-fi, horror, and adventure.

E. Threshold Optimization Analysis

The optimal thresholds t_j^* listed in Table VIII vary substantially across genres and models, ranging from 0.05 to 0.89 for Experiment I and 0.05 to 0.81 for Experiment II. This variability confirms that a uniform threshold of 0.5 is systematically suboptimal for imbalanced multilabel settings.

A clear pattern emerges: minority genres with few positive test instances tend to receive low optimal thresholds (e.g., *musical*: $t^* = 0.05$; *romance*: 0.06 for Experiment I, 0.23 for Experiment II; *crime*: 0.13). This reflects the model’s conservative sigmoid outputs for underrepresented classes, since positive-weight BCE incentivizes recall, the model produces low-confidence positive scores for minority genres, requiring a low decision boundary to recover any true positives.

Conversely, visually or lexically unambiguous genres tend to have higher optimal thresholds (*family*: 0.89; *fantasy*: 0.76

for Experiment I), indicating that the model produces high-confidence positive predictions for these genres but with poor precision, necessitating a higher threshold to suppress false positives.

For Experiment I, the macro-level gain from threshold optimization (+0.0791 Macro F1) is driven primarily by improvements on *action* (+0.0365), *animation* (+0.0762), *comedy* (+0.2231), *crime* (+0.1190), and *sci-fi* (+0.0344). For Experiment II, larger gains on *sci-fi* (+0.3200) and *thriller* (+0.0525) drive the overall improvement.

F. Discussion

The results highlight three principal findings.

Title information provides a large and consistent advantage. The multimodal architecture achieves substantially higher performance across all metrics and thresholds, with the gap most pronounced on genres with strong lexical title cues. The cross-attention mechanism enables selective exploitation of title semantics to guide visual feature interpretation, resulting in more precise per-genre predictions and better-calibrated sigmoid outputs. This confirms that movie titles, despite being a lightweight and universally available textual signal, encode genre-discriminative information that complements visual poster features in a non-trivial way.

Class imbalance is the primary performance bottleneck. The complete failure on *musical* and near-failure on *romance* and *family* underscores that severe class imbalance, particularly genres with fewer than 5 positive test samples, represents a fundamental limitation that no architectural innovation can overcome without sufficient training data. Targeted oversampling and positive-weight BCE reduce the false-negative rate on these genres during training but cannot generate new discriminative signal where none exists.

Model capacity must be matched to dataset size. The Phase 1 overfitting pattern and the persistent train-validation gap in Phase 2 of Experiment I illustrate the mismatch between the $\sim 124\text{M}$ -parameter multimodal model and a dataset of 250 training samples. While the multimodal model ultimately outperforms the image-only baseline, indicating that the additional parameters do carry useful inductive bias, the BERT component introduces a large regularization burden. Replacing BERT with a lighter encoder (e.g., DistilBERT or SentenceTransformers) could reduce this burden while retaining semantic title representations, making the multimodal approach more practical for small-scale datasets.

Future work should address these limitations through: (1) scaling training data with broader IMDb or TMDB poster collections to reduce class imbalance; (2) exploring asymmetric loss functions or label-smoothing for extreme minority classes; (3) substituting BERT-base with lightweight text encoders to better match model capacity to dataset size; and (4) investigating label dependency modeling (e.g., graph neural networks or label co-occurrence loss terms) to exploit the strong inter-genre correlations identified in the EDA phase.

VI. CONCLUSION AND FUTURE WORKS

A. Conclusion

This paper presented a controlled comparative study on multilabel movie genre classification from poster images, evaluating two architectures: an image-only EfficientNet-B3 baseline and a multimodal model that fuses poster visual features with BERT-encoded movie title embeddings via an 8-head cross-attention mechanism. Both configurations were trained and evaluated on a dataset of 300 IMDb poster images spanning 14 genre categories.

The multimodal architecture (Experiment I) consistently outperformed the image-only baseline (Experiment II) across all evaluation metrics. At the default threshold of $t = 0.5$, Experiment I achieved a Macro F1 of 0.6502 versus 0.4554 for Experiment II, with further improvement to 0.7293 versus 0.5269 after per-genre threshold optimization. The advantage was most pronounced on genres with strong lexical title cues, particularly *sci-fi*, *horror*, and *adventure*, where title embeddings guided the cross-attention mechanism to attend to genre-discriminative spatial regions that visual features alone could not reliably distinguish. GradCAM visualizations confirmed that the cross-attention fusion produces spatially structured, genre-specific activation patterns, providing mechanistic support for the quantitative performance gains.

Nevertheless, several limitations were identified. Extreme class sparsity rendered *musical* and *romance* practically unclassifiable regardless of architecture or threshold strategy. Furthermore, Phase 2 training dynamics revealed an overfitting pressure arising from the mismatch between the $\sim 124\text{M}$ -parameter multimodal model and the small 250-sample training set, suggesting that lighter text encoders may be more appropriate for this data regime. These findings collectively confirm that title-based textual context is a lightweight yet highly effective complement to visual poster features, and that class imbalance and dataset scale represent the primary performance bottlenecks for this task.

B. Future Works

Future work should address the identified limitations through the following directions. First, scaling the training corpus with broader IMDb or TMDB poster collections would reduce class imbalance and provide sufficient positive samples for severely underrepresented genres such as *musical* and *romance*, where no architectural improvement can compensate for the near-absence of training signal. Second, the strong inter-genre co-occurrence patterns identified in the EDA phase, such as the frequent pairing of *thriller* with *horror*, and the near-universal co-occurrence of *sci-fi* with *action* and *adventure*, motivate the development of models that explicitly capture label dependencies. Incorporating label co-occurrence graph neural networks or co-occurrence-aware loss terms could enable the classifier to leverage inter-genre correlations rather than treating each label as an independent binary prediction, potentially improving performance on correlated genre pairs where isolated predictions fail. Third, replacing

BERT-base with lightweight alternatives such as DistilBERT or SentenceTransformers would reduce the model-to-data ratio and the associated overfitting burden, making the multimodal approach more practical for small-scale datasets. Fourth, asymmetric loss functions or label-smoothing strategies tailored to extreme minority classes could further improve gradient signal beyond what positive-weight BCE and targeted oversampling can achieve. Finally, future work could investigate the incremental contribution of additional prerelease metadata, such as taglines, cast information, or release year, alongside movie titles, while preserving the practical constraint of prerelease availability.

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